

Part A - User and System Investigation (10 minutes)

- Determine the username of the account you are currently using.
whoami
- Display your UID, primary GID, and supplementary groups.
UID: id -u, GID: id -g, Groups: id -G
- Display all users currently logged in to the system.
who
- Determine what activities the currently logged-in users are performing.
w
- Examine */etc/passwd* and identify the following information for your current account:
Username
UID
GID
Home directory
Login shell

grep “^\$(whoami):” /etc/passwd

- Determine how many user-account entries are present in */etc/passwd*.
wc -l /etc/passwd
- Display only the username and UID fields from */etc/passwd*.
cut -d: -f1,3 /etc/passwd
- Find all accounts whose UID is 0. Explain the security significance of UID 0.
awk -F: '\$3 == 0 {print \$1}' /etc/passwd

Security significance of UID 0:

UID 0 represents the root/superuser account.

A user with UID 0 has essentially unrestricted access to the system.

An unauthorized UID 0 account is a serious security risk. Attackers who obtain UID 0 effectively have administrator-level control.

- Determine how many user-account entries are present in /etc/passwd.
`wc -l /etc/passwd`
- Display only the username and UID fields from /etc/passwd.
`cut -d: -f1,3 /etc/passwd`
- Find all accounts whose UID is 0. Explain the security significance of UID 0.
`awk -F: '$3 == 0 {print $1}' /etc/passwd`

Security significance of UID 0:

UID 0 represents the root/superuser account.

A user with UID 0 has essentially unrestricted access to the system.

An unauthorized UID 0 account is a serious security risk. Attackers who obtain UID 0 effectively have administrator-level control.

- Compare /etc/passwd and /etc/shadow. Why are password hashes stored in /etc/shadow instead of /etc/passwd?

/etc/passwd	/etc/shadow
Stores basic information about user accounts.	Stores password hashes and password-related information.
Contains username, UID, GID, home directory, and login shell.	Contains encrypted/hashed passwords and password-aging information.
It is generally readable by all users.	It has restricted permissions and is normally accessible only by root or authorized processes.
The password field usually contains x, indicating that the password hash is stored in /etc/shadow.	Contains the actual password hash.

Password hashes are stored in /etc/shadow because password hashes are sensitive information. The /etc/passwd file needs to be readable by ordinary users and system programs for user and account information. If password hashes were stored there, any user could access them and potentially perform offline password-cracking attacks.

Part B - Creating and Verifying Users (20 Minutes)

- Create the following users and ensure that a home directory is created for each:
alice, bob, Charlie
sudo useradd -m alice
sudo useradd -m bob
sudo useradd -m charlie
- Verify that all three accounts have been added to `/etc/passwd`.
grep -E '^(alice|bob|Charlie):' /etc/passwd
- Determine the UID assigned to each newly created user.
id -u
- Determine the home directory and default shell assigned to alice.
getent passwd alice

Home directory → /home/alice
Default shell → /bin/bash
- Inspect `/home` and verify that the required home directories were created.
ls -l /home
- Set passwords for all three accounts.
sudo passwd alice
- Switch from your current account to alice using a login shell. Verify the change using an appropriate command, and then return to your original account.
Switch from your current account to alice: *su -alice*
Verify the change: *whoami*, *pwd*, *id*
return to your original account: *exit*

Part C - Group Management and Access Control (30 Minutes)

- Create two groups:
security and developers
sudo groupadd security
sudo groupadd developers
- Add alice and charlie to the security group.
sudo usermod -aG security alice
sudo usermod -aG security charie
- Add bob to the developers group.
sudo usermod -aG developers bob
- Without switching accounts, verify the group membership of all three users.
id alice or groups alice

- Display the entries corresponding to security and developers from /etc/group.
grep -E '^(security|developers):' /etc/group
- Add bob to the security group without removing him from the developers group.
sudo usermod -aG security bob
- Verify that bob now belongs to both groups.
groups bob